

The analysis identified 512 suspicious authentication-related events in the Linux log dataset. These events included authentication failures, failed-password attempts, invalid-user attempts, and unknown-user activity. Splunk visualization showed several spikes in authentication failures over time, suggesting periods of automated or concentrated login activity. The most active source was 150.183.249.110, which generated 80 authentication attempts. Repeated attempts against privileged accounts such as root may indicate brute-force activity. Recommended actions include reviewing successful logins that occurred after the failures, disabling direct root SSH login, requiring SSH key authentication, applying rate limiting, blocking confirmed malicious sources, and continuing to monitor authentication logs.